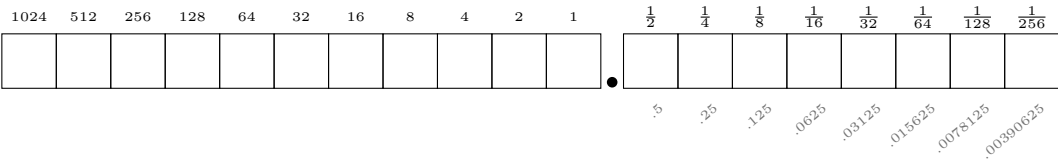


FP32 1/2 IEEE 754 single precision: 1 sign, 8 exponent (bias 127), 23 mantissa. value = $\pm 1.m \times 2^{E-127}$

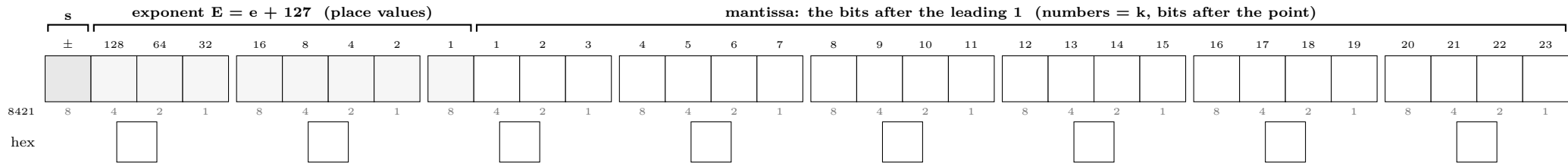
ENCODE decimal $\rightarrow$ hex

lecture: $-6.5 = \text{C0D00000}$

1 sign: negative $\Rightarrow$ s = 1. **2** write $|x|$ in binary (subtract the biggest place value that fits, repeat)



5 fill s, E, then the bits after the leading 1 (pad with 0s). **6** read each nibble with 8421 $\rightarrow$ hex.



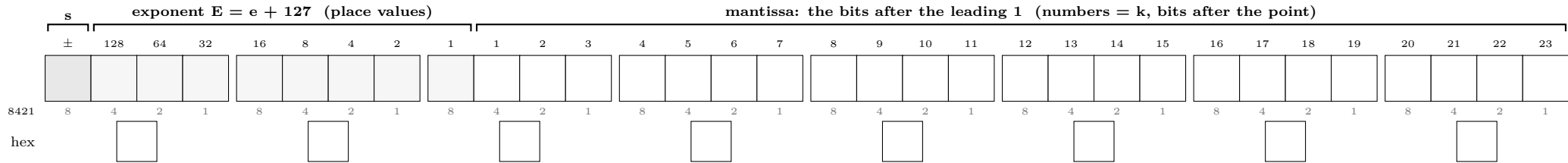
e.g. $85.125 = 1010101.001 = 1.010101001 \times 2^6$; $E = 133 = 10000101$; $01100\ 0010\ 11010\ 1010\ 0100\ 0000\ \dots = 42\text{AA}4000$ (23/24 S2 Q2). $-1.0 = 1.0 \times 2^0$, $E = 127 \rightarrow \text{BF}800000$ (25/26 Q5)

CHECK E FIRST	E	00000000	00000000	1 – 254	11111111	11111111
	mantissa	all 0	not all 0	anything	all 0	not all 0
	value	± 0	denormal $\pm 0.m \times 2^{-126} = \pm N \times 2^{-126-k}$	normal $\pm 1.m \times 2^{E-127}$	$\pm \infty$	NaN
	hex	00000000 / 80000000	00000001–007FFFFF	00800000–7F7FFFFF	7F800000 / FF800000	7F800001–7FFFFFFF

DECODE hex $\rightarrow$ decimal value = $\pm N \times 2^{e-k}$

$E = 00000000$ or $11111111?$ $\rightarrow$ table above

1 write each hex digit in the bottom boxes, then its 4 bits above (8421). **2** E = add the exponent place values; $e = E - 127 = \underline{\hspace{2cm}}$



3 k = number of the *last* 1 in the mantissa (0 if none).

4 $N = 1$ followed by mantissa bits 1 to k , read as a whole number:

$N = 1.\underline{\hspace{2cm}}2 = \underline{\hspace{2cm}}$ (**BIN** type it, **DEC**)

5 value = $\pm N \times 2^{e-k}$ ($e - k < 0$: divide by 2^{k-e} ; leave Base-N first, it drops fractions)

= $\pm \underline{\hspace{2cm}} \times 2^{\underline{\hspace{2cm}}} = \underline{\hspace{2cm}}$

Calc shortcut for 2: multiply by 2^k until it is a whole number, convert that, then put the point k places from the right.

e.g. $85.125 \times 8 = 681$ **DEC** $\rightarrow$ **BIN** 1010101001 $\rightarrow$ 1010101.001 ($k = 3$)

3 normalise: move the point to just after the first 1.

1. $\underline{\hspace{2cm}} \times 2^e$, $e = \underline{\hspace{2cm}}$ (moved left: $e > 0$; right: $e < 0$)

4 $E = e + 127 = \underline{\hspace{2cm}}$ $\rightarrow$ bits with 128 64 32 16 8 4 2 1. $e \geq 1$: 1 then $(e - 1)$ in 7 bits

k	1	2	3	4	5	6	7	8	9	10	11	12
2^k	2	4	8	16	32	64	128	256	512	1024	2048	4096
k	13	14	15	16	17	18	19	20	21	22	23	
2^k	8192	16384	32768	65536	131072	262144	524288	1048576	2097152	4194304	8388608	

e.g. 42570000: $E = 10000100 = 132$, $e = 5$; mantissa 1010111, $k = 7$; $N = 11010111 = 215$; $215 \times 2^{5-7} = 215 \div 4 = \mathbf{53.75}$ (23/24 S2 Q3)

e.g. BEA00000: s = 1; $E = 125$, $e = -2$; mantissa 01, $k = 2$; $N = 101 = 5$; $5 \times 2^{-4} = 5 \div 16 \rightarrow \mathbf{-0.3125}$ (24/25 Q5)

Denormal ($E = 0$): no leading 1 and $e = -126$: $N =$ mantissa bits 1 to k , value = $\pm N \times 2^{-126-k}$.

FP32 2/2 IEEE 754 single precision: 1 sign, 8 exponent (bias 127), 23 mantissa. value = $\pm 1.m \times 2^{E-127}$

SPECIAL special values and range (Lecture 9)

E	mantissa	value	hex (+ / -)
00000000	0	± 0	00000000 / 80000000
00000000	$\neq 0$	denormal: $\pm 0.m \times 2^{-126}$	00000001-007FFFFF
1-254	any	normal: $\pm 1.m \times 2^{E-127}$	
11111111	0	$\pm \infty$	7F800000 / FF800000
11111111	$\neq 0$	NaN (an error result)	7F800001-7FFFFFFF

Denormals: hidden bit is **0** and the exponent is fixed at $1 - 127 = -126$ (not -127). They exist to reach numbers smaller than the smallest normal.

E all 0s or all 1s = special; anything else = normal with a hidden 1

landmark	hex	value
smallest + denormal	00000001	$2^{-149} \approx 1.4 \times 10^{-45}$
largest denormal	007FFFFF	$(1 - 2^{-23}) \times 2^{-126}$
smallest + normal	00800000	$2^{-126} \approx 1.18 \times 10^{-38}$ (24/25 Q4)
largest normal	7F7FFFFF	$(2 - 2^{-23}) \times 2^{127} \approx 3.4 \times 10^{38}$
most negative	FF7FFFFF	$-(2 - 2^{-23}) \times 2^{127}$
1.0 / -1.0	3F800000 / BF800000	E = 127
0.5 / 2.0 / 4.0	3F000000 / 40000000 / 40800000	E + 1 = hex + 00800000

Overflow: beyond the largest. **Underflow:** nonzero but smaller than the smallest. **Range** = the set of values it can hold; **precision** = smallest gap between two of them.

PRECISION what survives, rounding, other formats (Lectures 3D, 9)

24 significant bits (hidden 1 + 23), about 7 decimal digits. Gap between neighbours = 2^{e-23} : relative, not fixed.

Is the integer X exact? n = number of bits in X (up to its leading 1).
 $n \leq 24$: exact. $n > 24$: the lowest $n - 24$ bits are lost, so exact only if they are all 0.
Integers are exact up to $2^{24} = 16\,777\,216$; **16 777 217** = $2^{24} + 1$ is the first that is not (24/25 Q6).

Does $f(X) = f(X+1)$? (round toward zero) Yes when $n > 24$ and the lost bits of X are *not* all 1s, so the +1 never carries into the kept bits.
e.g. 0120101C: $n = 25$, lose 1 bit (0) $\rightarrow$ same $\checkmark$. 25A2D87F: $n = 30$, lose 6 bits 111111 $\rightarrow$ carry, different. (25/26 Q6)

Rounding (Lecture 9): **to nearest** (default; a tie goes to the one whose last bit is 0), **toward 0** (chop), **toward $+\infty$** , **toward $-\infty$** . Hardware keeps 3 extra bits: guard, round, sticky (OR of everything after).
Every operation rounds, so $(A + B) + C \neq A + (B + C)$ in general: **no associativity**.

format	bits	exp	frac	bias	use
FP64	64	11	52	1023	scientific
FP32	32	8	23	127	general
BF16	16	8	7	127	AI training
FP16	16	5	10	15	graphics, AI
FP8 E4M3	8	4	3	7	AI weights
FP8 E5M2	8	5	2	15	AI gradients
FP4 E2M1	4	2	1	1	experimental

bias = $2^{\text{exp bits}-1} - 1$. Same ENCODE/DECODE steps, just a different bias and widths.

INT8 quantisation: scale = $\frac{2^{b-1} - 1}{\max |x|}$, $q = \text{round}(x \times \text{scale})$.
e.g. $\max |x| = 12.47 \rightarrow \text{scale} = 127/12.47 = 10.18$; $-7.6 \rightarrow -77$, $5.72 \rightarrow 58$.

Fixed point (k fraction bits): precision 2^{-k} ; in 2s complement $111110.11 = -1.25$.